AUSTRALIAN MATHS TRUST

2020 AMC

AUSTRALIAN MATHEMATICS COMPETITION

Upper Primary Years 5–6
(Australian school years)

CHINESE VERSION

NAME _____

TIME ALLOWED: 60 MINUTES

INSTRUCTIONS AND INFORMATION

General

1. Do not open the booklet until told to do so by your teacher.
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones are not permitted.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

The answer sheet

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder: You may sit this competition once, in one division only, or risk no score.

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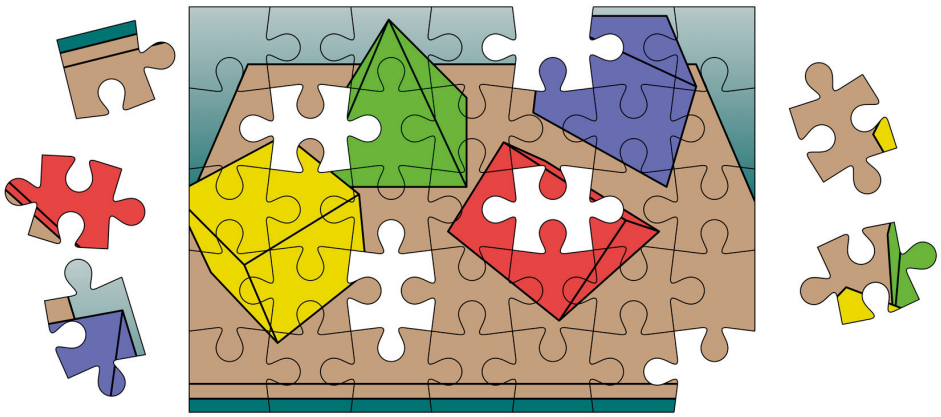
1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 截至目前为止，请问总共有几片拼图已经被排进图内？



How many pieces have been placed in the jigsaw puzzle so far?

(A) 25 (B) 27 (C) 30 (D) 33 (E) 35

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2. 请问2020 的一半是多少？

What is half of 2020?

(A) 20 (B) 101 (C) 110 (D) 1001 (E) 1010

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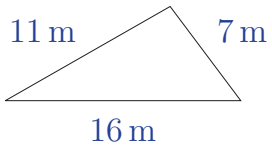
3. 请问右图的三角形周长是多少？



What is the perimeter of this triangle?

(A) 33 m (B) 34 m (C) 35 m (D) 36 m (E) 37 m

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4. 请问下列哪一个分数最大？

(A) 二分之一 (B) 四分之一 (C) 三分之一
(D) 四分之三 (E) 十分之六

Which fraction is the largest?

(A) one-half (B) one-quarter (C) one-third
(D) three-quarters (E) six-tenths

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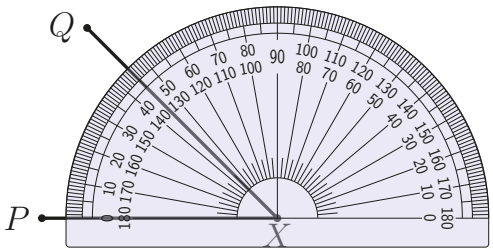
5. 用一个量角器丈量角 PXQ 的大小。
请问这个角的大小是多少度？

A protractor is used to measure angle PXQ.
The angle is



(A) 45°　(B) 55°　(C) 135°　(D) 145°　(E) 180°

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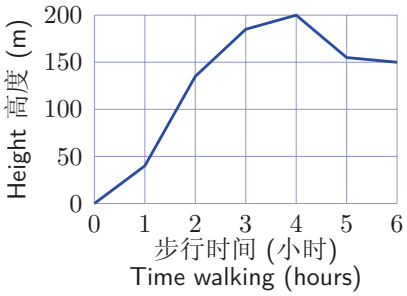
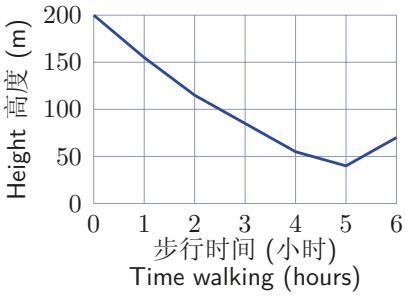
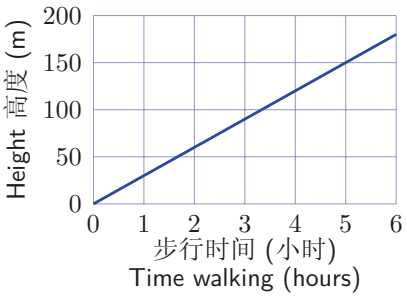
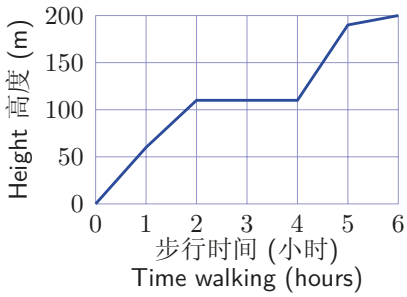
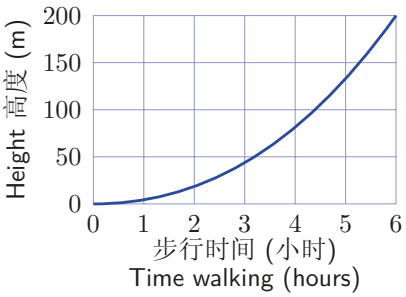
6. 一群朋友正朝着山里的一个湖前进。首先他们爬上一个山坡，然后向下走到这个湖。请问哪一个选项内的图最能正确表示他们的行程？
Some friends are walking to a lake in the mountains. First they climb a hill before they walk down to the lake. Which graph most accurately represents their journey?

(A)


(B)


(C)


(D)


(E)


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7. 请问多少个十分之一相加等于 6.2 ？

How many tenths are in 6.2?

(A) 62 (B) 8 (C) 4 (D) 12 (E) 36

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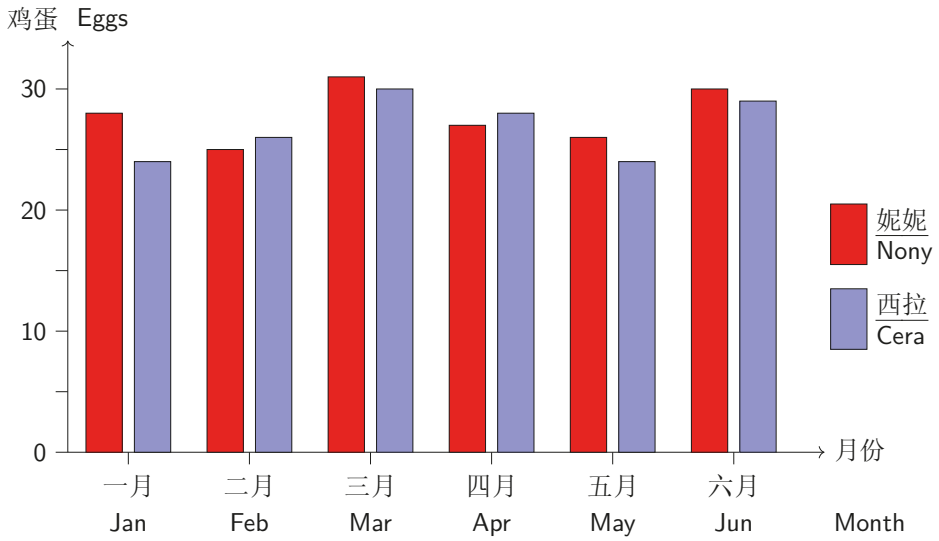
8. 后院饲养有两只鸡，妮妮与西拉。下图显示它们在今年前六个月所生出的鸡蛋数量。



请问有几个月妮妮生的蛋比西拉多？

The graph shows the number of eggs laid by backyard chickens Nony and Cera for the first six months of the year.

In how many months did Nony lay more eggs than Cera?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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9. 某班 24 名学生的身高各不相同，将他们从高到矮排成一列。从高到矮数起，玛莉排第八，从矮到高数起，约翰排第六。请问有多少名学生站在他们两人之间？

A class of 24 students, all of different heights, is standing in a line from tallest to shortest. Mary is the 8th tallest and John is the 6th shortest. How many students are standing between them in the line?

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

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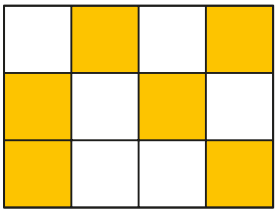
10. 小梅将一个矩形分割为若干个全等的小正方形，并将其中一部分涂色，如图所示。她希望将整个矩形面积的四分之三都涂色。请问她还需要将几个小正方形涂色？



Maria divided a rectangle into a number of identical squares and coloured some of them in, as shown.
She wants three-quarters of the rectangle's area to be coloured in altogether. How many more squares does she need to colour in?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 在弹珠游戏结束时，小李有 15 颗弹珠、小杜有 8 颗弹珠、小欧有 4 颗弹珠。小李应该一共还给他的朋友们多少颗弹珠，才能让下一局开始时大家的弹珠数量都相同？

At the end of a game of marbles, Lei has 15 marbles, Dora has 8 and Omar has 4. How many marbles must Lei give back to his friends if they want to start the next game with an equal number each?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

---

12. 在方格表中，每一行数字之和标在该行的最右边、每一列数字之和标在该列的最下边。请问 N 的值是多少？

8		6	16
	N		9
7		9	20
20	7	18	

In the grid, the total of each row is given at the end of the row, and the total of each column is given at the bottom of the column.
The value of N is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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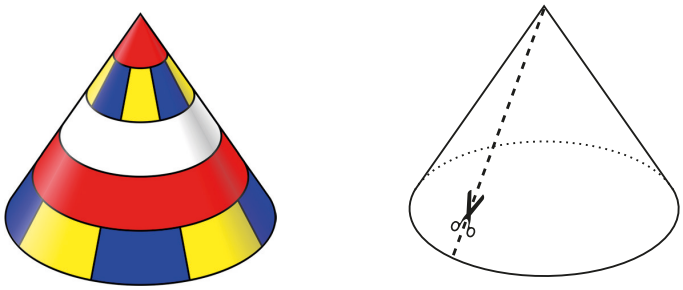
13. 在小力的生日宴会上，他与他的朋友们都戴上由彩带作成圆锥状的帽子，如下左图所示。当宴会结束后，小力将其中一顶帽子沿着一条直线向上剪开至顶点，如右下图所示。



当小力把剪开的帽子展开贴平在桌面上时，请问下列哪一项将与这顶帽子的外观符合？

At his birthday party, Ricky and his friends wear stripy paper hats in the shape of a cone, as shown on the left. After the party, Ricky makes a straight cut in one of the hats all the way up to the point at the top, as shown on the right.

Which of the following best matches what the hat will look like when Ricky flattens it out on the table?

(A)

(B)

(C)

(D)

(E)

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14. 小艾打算按照顺序将 1 到 50 依次写在纸上。她在第一行共写了25个数字。请问第一行的最后一个数是多少？

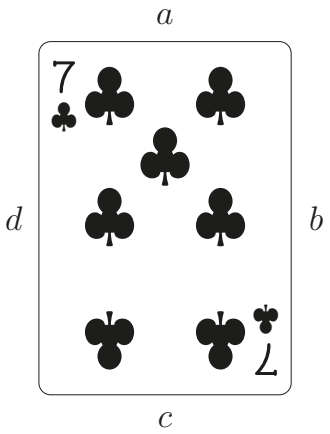
Emma is going to write all the numbers from 1 to 50 in order. She writes 25 digits on the first line of her page. What was the last number she wrote on this line?

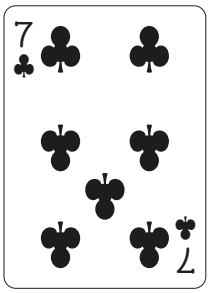
(A) 13 (B) 15 (C) 17 (D) 19 (E) 21

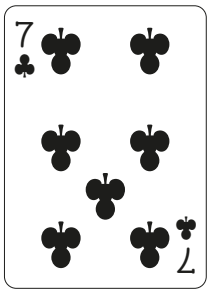
---

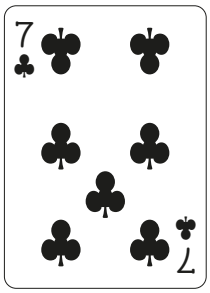
15. 将右图所示的扑克牌沿着 b 边翻转，然后再沿着 c 边翻转。
请问下列哪一项是它经过两次翻转之后的样子？

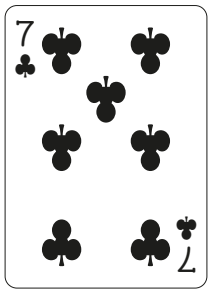
The playing card shown is flipped over along edge b and then flipped over again along edge c.
What does it look like now?

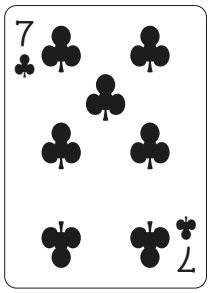


(A) 

(B) 

(C) 

(D) 

(E) 

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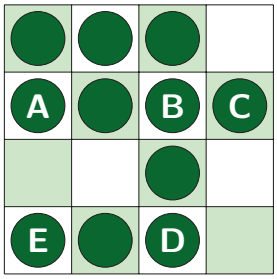
16. 请问要移除哪一个字母坐标，才能使每一行上的棋子个数都不相同且每一列上的棋子个数也不相同？

Which labelled counter should you remove so that no two rows have the same number of counters and no two columns have the same number of counters?



(A) A (B) B (C) C (D) D (E) E

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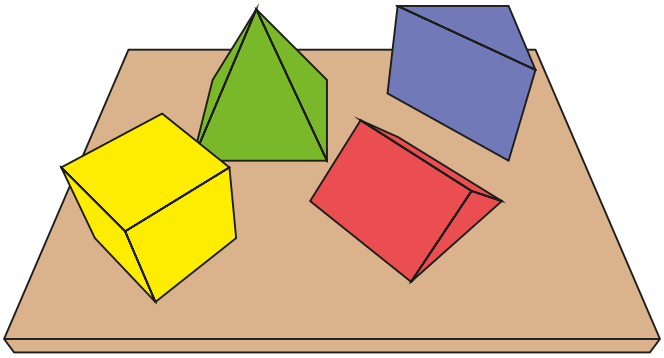
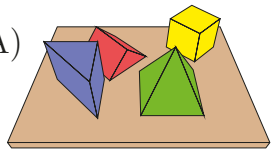
17. 阿丹在学校的书桌上放了各种形状的立体模型。下图是从他这一侧书桌看过去的样子：

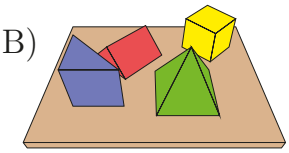


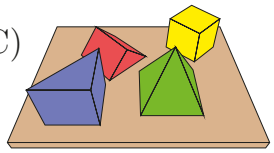
小南面对阿丹坐在桌子的对面。请问下列哪一项内的图形可表示从小南这一侧桌子看过去的图形？

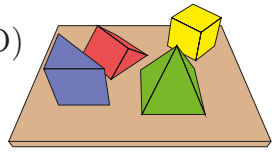
Aidan puts a range of 3D shapes on his desk at school. The diagram shows the view from his side of the desk.

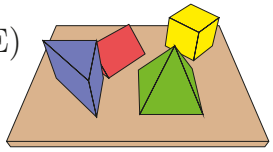
Nadia is sitting on the opposite side of the desk facing Aidan. Which of the following diagrams best represents the view from Nadia's side of the desk?

(A) 

(B) 

(C) 

(D) 

(E) 

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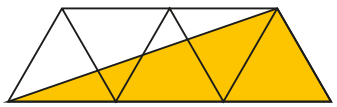
18. 右图中每一个小正三角形的面积都是 $1\,\text{m}^2$ 。请问阴影部分的面积是多少？



The area of each of the five equilateral triangles in the diagram is 1 square metre. What is the shaded area?

(A) $1.5\,\text{m}^2$ (B) $2\,\text{m}^2$ (C) $2.5\,\text{m}^2$ (D) $3\,\text{m}^2$ (E) $3.5\,\text{m}^2$

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19. 小凯今年 5 岁，小雷比小克小13 岁。一年前，小克的年龄是小凯与小雷年龄和的两倍。请问这三个孩子目前年龄的总和是多少岁？

Kayla is 5 years old and Ryan is 13 years younger than Cody. One year ago, Cody's age was twice the sum of Kayla's and Ryan's age. Find the sum of the three children's current ages.

(A) 10 (B) 22 (C) 26 (D) 30 (E) 36

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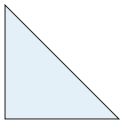
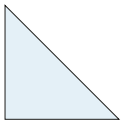
20. 玛莉将一张纸对折两次之后，最后得到以下形状：



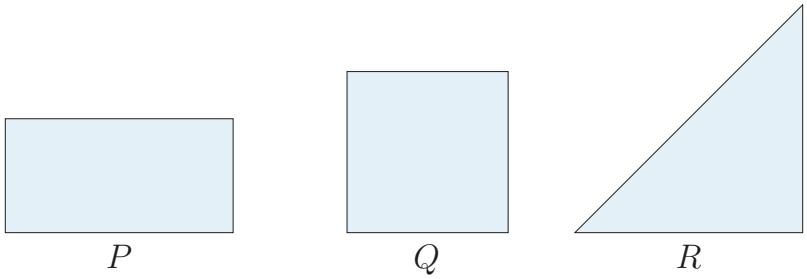
以下所示的形状 P、Q、R 中，请问哪些可能是它初始时的形状？

(A) 只有P (B) 只有Q (C) 只有R (D) 只有P与R (E) 三个都是

Mary has a piece of paper. She folds it exactly in half. Then she folds it in half again. She finishes up with this shape.



Which of the shapes P, Q and R shown below could have been her starting shape?



(A) only P (B) only Q (C) only R (D) only P and R (E) all three

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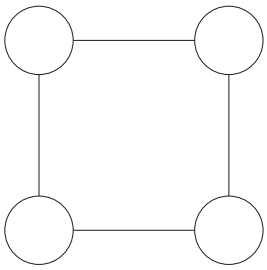
21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 将四个正整数填在一个正方形的四个顶点上，如图所示。对于每一条边，将两个顶点上的数字之差写在这条边上。已知四条边上的数分别为 1、2、3、4。请问顶点上四个数之和的最小可能值是多少？



Four positive whole numbers are placed at the vertices of a square. On each edge, the difference between the two numbers at the vertices is written. The four edge numbers are 1, 2, 3 and 4 in some order. What is the smallest possible sum of the numbers at the vertices?

(A) 10 (B) 11 (C) 12 (D) 13 (E) 14

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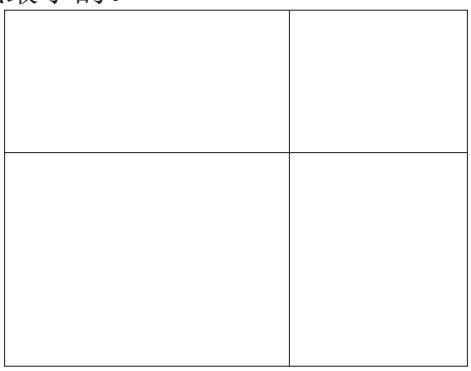
22. 将下图所示的大矩形分割成四个小矩形，已知其中三个小矩形的周长分别为10 cm、16 cm 与 20 cm，第四个小矩形的周长不是这四个小矩形中最大或最小的。



请问这个大矩形的周长是多少 cm？

The large rectangle shown has been divided into 4 smaller rectangles. The perimeters of three of these are 10 cm, 16 cm and 20 cm. The fourth rectangle does not have the largest or the smallest perimeter of the four smaller rectangles.

What, in centimetres, is the perimeter of the large rectangle?

(A) 26 (B) 30 (C) 32 (D) 36 (E) 46

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23. 一捆草可以供一匹马吃 2 天、供一头牛吃 3 天、供一只羊吃 12 天。有一位农夫用 22 捆草来喂一匹马、一头牛与一只羊。请问这些草可以维持多少天？

A bale of hay can be eaten by a horse in 2 days, by a cow in 3 days and by a sheep in 12 days. A farmer has 22 bales of hay and one horse, one cow and one sheep to feed. How many days will his bales last?

(A) 20 (B) 22 (C) 24 (D) 26 (E) 28

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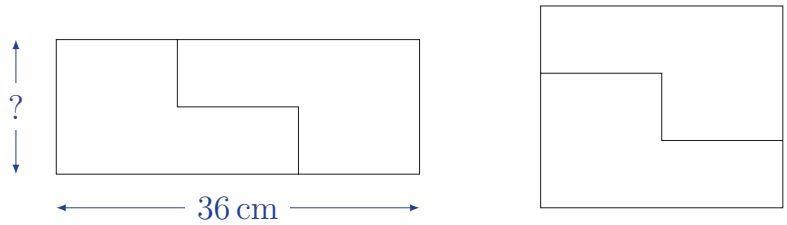
24. 下图这个矩形的长边为 36 cm。将它分割成两部分后重组，可以组成一个正方形，如图所示。



请问原来矩形的宽是多少？

This rectangle is 36 cm long. It is cut into two pieces and rearranged to form a square, as shown.

What is the height of the original rectangle?

(A) 14 cm (B) 16 cm (C) 18 cm (D) 20 cm (E) 24 cm

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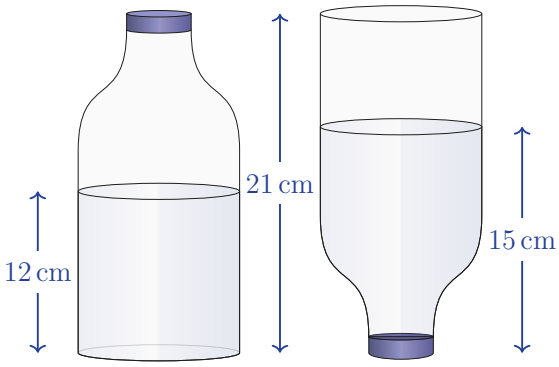
25. 右图显示了一个装有水的密封瓶子正立放置与倒立放置时的情况。请问这个瓶子中装了多少体积的水？

A bottle with a sealed lid contains some water. The diagram shows this bottle up the right way and upside down.
How full is the bottle?



(A) $\frac{1}{2}$ (B) $\frac{4}{7}$ (C) $\frac{5}{7}$ (D) $\frac{2}{3}$ (E) $\frac{9}{14}$

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 如果一个数各位上的数字都是奇数，则我们称这个数为“奇异数”。例如，9、57、313 都是奇异数。
但是 50 与 787 则不是奇异数，因为 0 与 8 都不是奇数。
请问从 1 到 999 总共有多少个数是奇异数？

A number is oddtastic if all of its digits are odd.
For example, 9, 57 and 313 are oddtastic.
However, 50 and 787 are not oddtastic, since 0 and 8 are even digits.
How many of the numbers from 1 to 999 are oddtastic?

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27. 我的养鸡场有24个栅栏，每个栅栏内都有相同数量的鸡。由于栅栏内有些拥挤，所以我新建了6个栅栏，这使得每个栅栏内鸡的数量都减少了 6 只。请问我总共有多少只鸡？

On my chicken farm where I have 24 pens, the pens were a bit crowded. So I built 6 more pens, and the number of chickens in each pen reduced by 6. How many chickens do I have?

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28. 各位数字之和等于8，且是偶数的三位数共有多少个？

How many even three-digit numbers are there where the digits add up to 8?

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29. 小美用这个键盘输入她的三位验证码 (PIN)。这三个数字都互不相同，且第一个数字与第二个数字的按键相邻、第二个数字与第三个数字的按键相邻。

例如，563 是一个可能的验证码，但是 536 则不是，因为按键 5 与 3 不相邻。
请问小美输入的验证码有多少种可能性？

Madeleine types her three-digit Personal Identification Number (PIN) into this keypad.

All three digits are different, but the buttons for the first and second digits share an edge, and the buttons for the second and third digits share an edge.

For instance, 563 is a possible PIN, but 536 is not, since 5 and 3 do not share an edge.

How many possibilities are there for Madeleine's PIN?

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30. 写下一个数字需要一秒。给你半小时，从 1、2、3、$\cdots$ 开始，写出尽可能多的正整数。当半小时结束时，请问你刚写完哪一个数？

Writing one digit every second, you have half an hour to list as many of the counting numbers as you can, starting 1, 2, 3, … . At the end of half an hour, what number have you just finished writing?